

Student academic performance prediction based on machine learning techniques with feature importance and visual analytics

Thesis report

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Chapter 1 – Introduction

1.1. Background of study

Student academic performance is one of most concerned factors in educational institutions. The major issue in education is analysing the outcomes of the student's performance and identify what is influencing them. Students may face challenges that affect their capacity to learn, which results in their academic performance. Analysing the academic data of the student in advance will allow early support decisions on their performance [1].

The growth of virtual learning environments has made ways for educational institutions to analyse and understand student performance. The advancement of learning Analytics (LA) and Educational Data Mining (EDM) paved way for analysing not only student's academic performance but also behaviour and engagement which helps the educational institutions to make decision by understanding the learning trends and predict student success rate [2], [3].

Many studies used machine learning techniques such as Decision Tree, Random Forest and Linear Regression models for prediction of student academic outcome. The focus on improving the accuracy of the performance prediction is achieved with the machine learning techniques and educational data mining techniques [4], [5]. As a result, these recent studies deployed machine learning models as black box systems rather than interpretability which makes it challenging for educational institutions to understand the aspects that influence the success of the students. Moreover, the predicted results are often represented as statistical evaluation metrics rather than providing easy to understand visual insights [6], [7].

This research study aims to predict the Global Score by analysing the student academic performance through data collection and preprocessing, exploratory data analysis, machine learning model implementation with appropriate evaluation metrics, integration of explainable AI technique SHAP and visualisation. This enables the educational institutions to analyse, predict and prioritize appropriate actions [8], [9]

1.2.Literature Review

The literature reviewed is organized in a structure as relevance tree. The following Fig. 1 represents the structure with each of the research papers being reviewed. The study mainly falls under four branches. They are: Exploratory Data Analysis, Algorithms, Interpretability and Learning Analytics Dashboards (LAD). Each article will be examined and discussed in the subsequent sections.

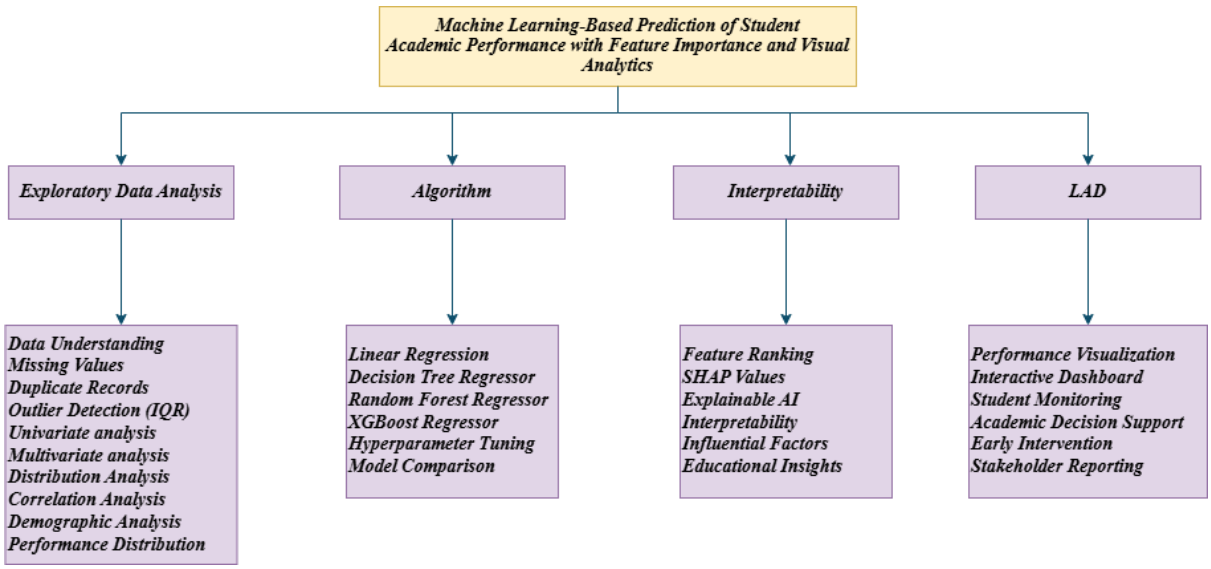


Figure 1. Relevance tree of literature review.

1.2.1. Prediction of student performance in an Electromagnetics course

[10] investigated the application of machine learning approaches to predict student academic performance in an Electromagnetics course and examined behavioural indicators resourced from Student Led Tutorial sessions. Two primary data have been used: students' preparation represented as ticks and other emerged from tutor student voice recordings. Using KNN approach with 5-fold cross validation, the student results were sorted as fail or pass category based on their participation pattern. The overall findings revealed ticks has a positive correlation between number of tutorial exercises and final grades with p value ($p = 0.618141$) while the voice data was a weak predictor for the academic outcome. The prediction strength has reached 65% accuracy. A key strength is the use of behavioural analytics instead of demographic or historical data. Though the observations of real time

data identify the students at risk in academics, the study has its limitation with the voice data and the usage of only one machine learning algorithm. Moreover, there is no specific techniques used for feature importance analysis to improve interpretability[10]

1.2.2. XGBoost based student performance prediction feature importance analysis

[11] proposed a system using XGBoost to predict how students perform across different areas along with feature important analysis. The study aimed to improve academic decision making by identifying the meaningful factors that strongly influence the results and predict academic outcome. From the data used, the indicators such as homework scores, class participation, attendance rate, test performance were utilized to build a predictive framework. The study aimed to impose techniques to analyse important features that has impact on each attribute. XGBoost algorithm was used to build a predictive framework. To evaluate and enhance generalizability and overcome overfitting, the model is trained with hyperparameters and 5-fold cross validation. The study result showed that XGBoost outperformed KNN algorithm by achieving 92% of prediction accuracy while KNN achieved 87% accuracy. Furthermore, homework scores and class engagement were the strongest factors of academic success rate among the academic features while attendance and test scores were weakest influential features according to the experimental results of feature importance analysis.

One major strength in the research is combining predictive model with feature importance analysis that identifies key features so insights into influential factors of learning outcomes alongside accurate prediction. The experimental outcome proved that the academic performance of students not only rely on examination results but highlights the importance of continuous student engagement indicators. Moreover, the research emphasises the utilization of XGBoost algorithm and its effectiveness in academic data mining applications. The limitations observed here is that the data size and the use of limited academic features which affects the model generalisability. Furthermore, the study lacked advanced XAI techniques for more interpretability but relied on the model's built in feature importance technique [11].

1.2.3. Prediction of student performance in higher education

[12] built a predictive system grounded in machine learning to assess how students from higher education perform in online education platforms particularly during the COVID -19 pandemic. The proposed methodology was applied on the Kaggle student performance dataset which contains 480 records and 17 features and then later trained and tested the dataset by dividing in 60:40 ratio. The study incorporated feature selection technique by performing BestFirst search method. The machine learning algorithm utilized in this article were K*, support vector machine, random forest and Naive Bayes. Recall, precision and F-measure metrics evaluated the efficiency of each of the model. The experimental findings demonstrated that the J48 classifier showed stronger results than the competing algorithms by revealing highest evaluation values. This paper focused on identifying at below average, average and advanced performers and improving student outcomes by enabling educators to provide targeted support using the machine learning models. Though the machine learning models identified academic success of the learners with desirable outcome, the reliance on the small dataset restricted the broader conclusion. Exploring more advanced algorithms on richer data source may benefit the educators.

1.2.4. Multiclass model for student grade prediction

[13] proposed a multiclass prediction model for student grade prediction. The class imbalance is addressed before model training with the use of SMOTE along with feature selection techniques. The SMOTE is synthetic minority oversampling technique, a statistical approach used to deal class imbalance by oversampling samples in datasets. The author has utilized “six machine learning techniques - Decision Tree (J48), Support Vector Machine (SVM), Naïve Bayes (NB), K-Nearest Neighbour (KNN), Logistic Regression (LR) and Random Forest (RF) using 1282 real student's course grade dataset”. In order to yield appropriate results, the predictive model was assessed with classification accuracy, precision, recall, and F-measure. Despite the imbalance, the SMOTE has gained the predictive performance over all the classifiers. The Random Forest model achieved the highest accuracy with 99.5% marginally better than J48 and KNN. Overall, the study reveals the effectiveness of the combination of oversampling and feature selection techniques to enhance multiclass student grade prediction [13].

1.2.5. Student academic performance using educational data mining

[14] proposed student academic performance prediction approach by assessing academic and practical performances of the student's characteristics. The study proposing the Relationship Matrix-based bipartite network approach, Relationship Matrix Hybrid Neural Network model, and with Louvain clustering as a solution to the problem of identifying relationships between sample attributes when studying multidimensional data. Louvain clustering was used to overcome the difficulty faced by the k-means model for grouping the students based on academic performance. The authors considered both the common characteristics and the individual traits of each student from different subsets. Accuracy, recall, F1 score, and precision were used to assess the performance of the built framework. The analysis of experimental results showed that the highest accuracy obtained in GCN when scale is 0.0695 and GAT at scale 0.075. The application of normalization to RMHNN improved the “accuracy and F1 score of models in predicting student performance”. The RMHNN is compared with the traditional approaches and achieved an accuracy of 93.10% with an F1 score above 90% [14].

1.2.6. Student academic performance prediction using predictive models

[15] aimed to propose a predictive machine learning-based framework supporting educational decision making by identifying the most effective algorithm using UCI Student Performance dataset. The authors experimented with several machine learning algorithms including logistic regression, Random Forest, Support Vector Machine, and AdaBoost. Prior to model training and evaluation, data underwent preprocessing and transformation techniques. Later in order to evaluate the models built, accuracy, precision, F1 score metrics were used. The experimental findings demonstrated that among all the algorithms used Random Forest achieved the highest prediction accuracy. Overall, the study revealed that the final grade of the student G3 had a significantly influenced by the previous scores from G1 AND G2 which suggests that historical academic records are the strongest indicators of the upcoming final academic success. Though the study utilized the potential of Random Forest algorithm for demonstrating its effectiveness in educational decision making, the research lacked the more advanced techniques.

1.2.7. Literature review on machine learning-based prediction of academic performance of students

The authors worked on 30 data driven research papers from the year 2015 to 2019. The aim of this research paper is to examine the recent machine learning methods to predict the performance of students from 2015 to 2019. The study found that predictive model can be accurate by using regression, classification and clustering techniques but classification is most commonly used technology to predict students' academic performance. These include decision tree, artificial neural network (ANN), k-nearest neighbour (KNN), Naive Bayes, and support vector machine (SVM), The authors concluded that there has been substantial rise in research on this area recently. The most significant academic factors were CGPA and attendance. The other influential factors contributing to the prediction accuracy includes gender, quiz, assignment and personal attributes. With 98.3% accuracy, ANN produced the best performance. The second highest accuracy with 98.2% decision tree while K-nearest neighbour and support vector machine achieved 95.8% and 91.3% respectively. The Naïve Bayes model produced the lowest accuracy [16].

1.2.8. Student performance prediction using feature importance analysis, Random Forest and SHAP

According to relevance tree, the concept of explainable artificial intelligence (XAI) was introduced in student academic prediction by [17]. The study explored feature importance analysis by a comparative analysis using Random Forest and Artificial Neural Network using Shapley Additive Explanations (SHAP) technique. The authors employed OULAD dataset containing 32000 students records of distance learning courses. The final dataset includes variables describing demographic characteristics (e.g., age band, gender, socio-economic indicators), assessment-related features (e.g., number of assessments, submission behaviour, score statistics, and assessment delays), and behavioural engagement features (e.g., total clicks, active days, intensity of activity, and resource-specific engagement).

After dealing with data preprocessing which includes data integration, cleaning and imputation, feature aggregation and encoding, the authors applied two machine learning models, RF and ANN to predict student academic performance. The RF classifier implemented with 200 estimators, balanced class weights and random seed of 42 for binary prediction task whereas the ANN captured complex interactions among the demographic,

academic and behavioural by utilising feedforward neural network with ReLU activation and batch normalization. Though ANN shows competitive result, it limits transparency in educational applications. SHAP was used along with ANN model to overcome this limitation.

The experimental results demonstrated that RF achieved 1.00 accuracy in training and maintained 0.93 accuracy in testing and ANN also achieved 0.93 accuracy both resulting in good generalization. The findings of feature importance with RF said that number of assessments, score statistics and interaction features were the most influential factors. The application of SHAP resulted in highlighting broader range of engagement features including activity specific interaction measures. Rather than ANN's nuanced patterns result. Overall, the study achieved the evaluation of predictive performance to ensure RF and ANN provided fair comparison of explainability methods.

1.2.9. Literature review on explainable artificial intelligence

According to the relevance tree, the research article by [18] conducted a literature review on the emerging concept XAI, Explainable Artificial Intelligence. The research paper focuses on the techniques applications and the challenges using the PRISMA methodology to review the research articles published between 2016 and 2024. The difference between interpretability and transparency is important in the advancement of XAI. Methods that produce explanations that focus on the on the feature importance are LIME, SHAP etc.

The article contributed its research on XAI applications on various fields including education, health care, finance, cybersecurity, agriculture and environmental science. The authors revealed that there were domain specific challenges that the XAI solutions must be designed to meet the norms and specification of various industry because each domain had unique requirements and conditions ranging from data privacy to regulations. The study also demonstrated that there were challenges despite the advancements such as the reduction of computational issue with some XAI techniques and balancing interpretability and high performance of complex models. The study concludes that XAI is an important factor in transparency, reliability and user focused AI systems.

1.2.10. Learning Analytics Dashboard in student academics

Nakaya and team proposed this research paper where they implemented the concept of Learning Analytics Dashboards (LADs) as an important tool for career planning and self-learning. LAD can be done in two approaches to students who explore the micro perspective at the course and improve self-regulated learning. The authors considered two approaches. The first is a top-down technique that analyses the support according to educational concepts and then the bottom-up strategy that explores the need of the students. This study aimed to adopt Participatory Design Research (PDR) to focus on students' preferences. This research paper found that the private sector category focused on self-learning while the graduate school category focused on learning from others, seeking help from others for career options and finally the study abroad category focused on analysis on requirements, scholarships, career chances, and alumni experience. According to classification of features based on profession as target, students who have defined career aim often wanted to follow the path of alumni with similar professional choice for better guidance while the students with no specific professional goal would be influenced by the career choice of peers with similar interest [19].

1.3. Problem Statement

A review of numerous existing researches has shown that machine learning models and educational data mining techniques are capable of predicting student performance, many studies focused on increased prediction accuracy. Thus, these prior researches often used models as black -box systems rather than interpretability, which makes it challenging for teachers to understand the contributing factors that affect the student's success in academics. Also, the existing models perform predictions where target data leakage can affect prediction.

Furthermore, the outcomes of the predictions are frequently displayed as statistical evaluation measures rather than providing an easy-to-understand visual interpretable insights.

1.4. Research Question

- Which machine learning model performed the best in terms of predictive accuracy and how can interpretability be incorporated with machine learning models to identify the factors most influencing the student performance?
- How do visual analytical dashboards enable educators improve the understanding of prediction?

1.5. Aim of the study

This proposed paper aims to address these gaps with the development of effective predictive model that combines machine learning techniques with feature importance analysis and visualisation techniques to improve interpretability and present insights from data visualization for educators to predict at-risk students early and support their outcomes. The proposed approach compares the outcome of four regression machine learning models such as linear regression, random forest, decision tree and extreme gradient boosting and identify the most important factors affecting student performance.

1.6. Objective

- To implement exploratory data analysis on the student's data.
- To build a leakage aware regression pipeline.
- To build machine learning models to predict students' global score.
- To use feature importance analysis to determine important affecting factors.
- To evaluate the performance of build model and interpretability.
- To create visual analytics dashboards that are interactive and provide actionable insights to the educators.

Chapter 2 – Research Methodology

The research methodology chapter focus on the systematic predictive framework. The methodology is implemented in sequence of optimized steps from processing the target dataset, data preparation, exploratory data analysis, model training, evaluation and comparison, interpretability metrics and visual analytics. The structure of this pipeline is illustrated in Fig

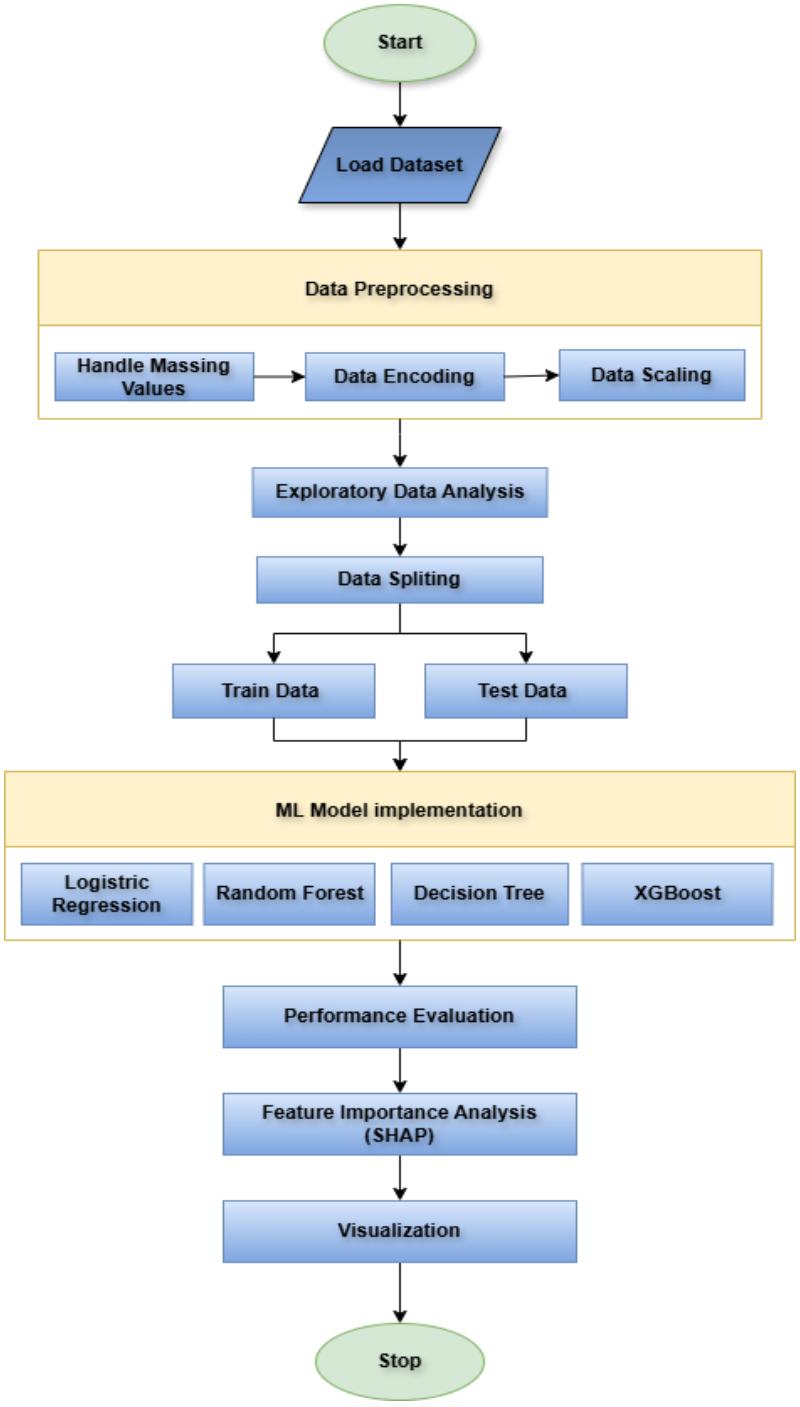


Figure 2. Flow chart of Research Methodology.

2.1. Data Definition

The first step in the research methodology is loading the dataset. The data source utilized in this study was retrieved from the Mendeley Data Repository. The data were originally obtained by "collection, adaptation and adjustment of information from the evaluations carried out by the Colombian Institute for the Evaluation of Education (ICFES)", [20]. This dataset presents observations of 12,411 students with 44 variables. The observations present the personal information, socioeconomic status and academic assessment results of the Engineering students in Colombia. The academic assessment results were captured at two moments - first through the high school exams (SABER 11) which evaluates five modules and second through professional examination (SABER PRO) evaluating six generic academic modules. The reason for choosing this dataset is that these observations help to build tools contributing to the educational process since it allows specific impact of each variable. Moreover, social variables are helpful for better understanding of how socioeconomic factors impact the final results of the students. The dataset is available in a free to use, publicly available research dataset, Mendeley Data Repository which is an academic data platform and can accessed with the link - https://data.mendeley.com/datasets/83tcx8psxv/1 [20].

TABLE I. VARIABLES OF SABER11 VS SABERPRO.

SABER 11		SABER PRO	
VARIABLE	DESCRIPTION	VARIABLE	DESCRIPTION
MAT_S11	Mathematics	QR_PRO	Quantitative Reasoning
CR_S11	Critical Reading	CR_PRO	Critical Reading
CC_S11	Citizen Competencies	CC_PRO	Citizen Competencies
BIO_S11	Biology	ENG_PRO	English
ENG_S11	English	WC_PRO	Written Communication
		FEP_PRO	Formulation of Engineering Projects
		G_SC	Global Score

2.2. Data Preprocessing

Data preprocessing is the process that prepares the data for subsequent processing. The process involves cleaning inconsistencies to improve data quality, organizing raw data for better model performance and support better analysis and insights. The raw data may contain missing values, inconsistences, mistakes and other issues that might affect the accuracy of the experiment [21]. The data preprocessing step helps to overcome these difficulties and has been followed with exploratory data analysis to analyse and spot the hidden correlation and pattern before model implementation.

In order to enhance the data quality and make it ready for machine learning model, data preprocessing steps include:

- **Data cleaning:** removing redundant and inconsistent records or columns.
- **Handling missing data:** Missing data can impact the data integrity and subsequent processes. This can be addressed by removing missing vales or imputation methods. In the recent research paper, "the choice of imputation method depended on the nature of the data. For numeric variables, mean or median imputation has been employed, while for categorical variables, mode imputation has been used" [21].

  The mean imputation uses the following arithmetic mean formula:

$$x = \frac{1}{n}\sum_{i=1}^{n} x_i \quad (1)$$

  Where x_i is ith value

  n is the total number of values

  $\sum$ is the summation of values

  The median imputation technique is as follows:

  if n is odd [21],

$$\text{med}(x) = x_{\frac{n+1}{2}} \quad (2)$$

if n is even,

$$\text{med}(x) = \frac{x_{\left(\frac{n}{2}\right)} + x_{\left(\left(\frac{n}{2}\right)+1\right)}}{2} \quad (3)$$

- **Categorical variable encoding:** The following table presents the description of all the categorical variables of the dataset.

TABLE II. DESCRIPTION OF CATEGORICAL.

VARIABLE	DESCRIPTION
GENDER	Biological sex of the student
EDU_FATHER	Educational qualification by father
EDU_MOTHER	Educational qualification by mother
OCC_FATHER	Employment of the father
OCC_MOTHER	Employment of the mother
STRATUM	Social and economic classification of household
SISBEN	Socioeconomic classification of household
PEOPLE_HOUSE	Number of people in the house
INTERNET	Internet availability
TV	Television availability
COMPUTER	Computer availability
WASHING_MCH	Washing machine availability
MIC_OVEN	Microwave oven availability
CAR	Car availability
DVD	DVD availability
FRESH	Fresh
PHONE	Phone availability
MOBILE	Mobile availability
REVENUE	Family income category of the students

JOB	Student's job status
SCHOOL_NAME	Name of the school
SCHOOL_NAT	Nature of the school
SCHOOL_TYPE	Type of the school
COD_SPRO	Code SABER PRO
UNIVERSITY	University name
ACADEMIC_PROGRAM	Academic Program chosen by student
COD_S11	Code SABER 11

Categorical data is an inherent component of this dataset where some of the categorical values need to be converted into numerical format to allow mathematical and statistical analysis. The categorical variables are encoded based on type of category. Ordinal type of variables is assigned with ranking based technique while label encoding assigns each distinct category an integer and the nominal variables are meant to be handled with one hot encoding technique [20], [21].

- **Feature scaling:** Standardization of numerical variables with Standard Scalar.

2.3. Exploratory Data Analysis

Exploratory data analysis is a critical step in this study in order to conduct initial investigation in discovering dataset's patterns and missing data landscape which helps in determining how to handle data in subsequent analysis steps. The key reasons to do the Exploratory data analysis means:

- Identify data errors.
- Verification of assumptions.
- **Outlier detection:** Outliers are data entries that deviates abnormally from typical patterns which might significantly affect data quality, model training and its performance. So, handling outliers is an essential step in data preparation.
- Selection of suitable initial models or techniques.
- Evaluating the direction and relationships between explanatory and target feature.
- Determining how each variable relates across the explanatory variables [22].

It is not easy for a human to spot the important features in the dataset when the data volume is huge. This process helps in this scenario by finding trends, spot anomalies, test hypothesis and check assumptions using statistical as well as graphical techniques. Exploratory data analysis can be univariate or multivariate.

Univariate analysis:

As its name suggests, univariate analysis focus on only one dependable variable. This analysis can be done numerical as well as categorical variable type. The goal of univariate analysis is to inspect and define the raw data with summary statistics and examine the pattern. The central tendency including mean, median and mode, range, variance, standard deviation and quartiles can be can be discovered using univariate analysis [23]. It includes the plotting techniques like boxplots and histograms.

Multivariate analysis:

Multivariate analysis is used when there is a requirement to analyse more than two variables simultaneously especially used to assess the relationship of each variable in the complex dataset with the target variable [24]. This can be done using the scatter plots and Bar charts. Depending on the data type and the target variable, technique of multivariate analysis is decided such as cluster analysis, multiple regression analysis, factor analysis and principal component analysis.

2.4. Machine Learning Model Deployment

The dataset and the research problem associated with the student academic performance revealed that machine learning models should be a regression task. Regression model is used as a predictive statistical model that analyses the relationship between the independent exploratory variables and the dependent continuous target variable. Regression models are used in systems that does time series modelling, forecasting and predicting. This study suggests using four regression models including Linear Regression, Random Forest, Decision Trees and XGBoost. Model training and evaluation will be implemented by splitting the dataset into two; 80% of the data will be utilized to train the model and the remaining 20% for testing purpose.

2.4.1. Linear Regression

The first approach to regression model is linear regression which determines the statistical evaluation between a target variable and exploratory variables. The model determines the strength of predictors, interpret results and trend forecasting. In this model, the value of the target variable (y) is totally determined by the independent variable (x). The increase in x variable is directly proportional to the y variable which is mathematically expressed as below.

$$y = \beta 0 + \beta 1x \quad (4)$$

This is called positive relationship. In contrast, the negative relationship occurs when the y variable decreased because of the rise of x variable which can be expressed as

$$y = \beta 0 - \beta 1x \quad (5)$$

In this predictive framework, the coefficients β0 and β1are the intercept and slope gradient of the line of regression. The best match line is determined by calculating the values of β0 and β1 parameters. The line of regression is shaped by the predicted observations. Next the errors for each actual observation are computed. The final regression model is then defined with regression line that minimizes the errors. Reducing residual errors is the primary goal of the linear regression model. The accuracy of linear regression is measured using R squared(R2) and Root Mean Squared Error measures the linear regression accuracy (RMSE) [12], [15].

2.4.2. Decision Tree

As a supervised machine learning model, decision tree regressor is a which is used as a regression model to predict numerical outcomes continuously. The model continuously splits the dataset into subsets based on values and attributes at each tree node and creates a binary tree structure. This approach will repeatedly follow minimizing the variance of the target variable in the divided subsets until an appropriate stopping point is reached with given criteria like minimum sample split, maximum tree depth, minimum leaf node samples. The leaf is the final end node of the tree holding a predicted value typically the mean of the target values from the samples. from the samples. It handles categorical as well as numerical data and also identify the nonlinear correlation between descriptors and target

variable. Without regularization, it will overfit the training set and performance will be poor on unknown data [25].

2.4.3. Random Forest

Random Forest is a supervised machine learning algorithm that can be implemented in both regression and classification tasks. The RF leverages ensemble learning by combining predictions of different models to improve accuracy. This method falls under the category of bagging techniques, distinct from boosting. With the subsets of training data, bagging entails the creation of decision trees, allowing trees to run concurrently without interaction. The prediction accuracy is improved by multiple decision trees in Random Forest model. While training the data, the algorithm generates many decision trees using replacement based samples randomly to estimate regression tree averages [25]. This variation in training subsets assists trees in capturing diverse data patterns effectively. During the prediction phase, each tree independently generates its prediction [26]. In the context of regression, the output of each tree is a predicted value. Ultimately, the final prediction of the Random Forest model aggregates individual tree predictions, often using methods like mode or mean [12]. One of the advantages is that the large dataset with high dimensional complexity can be handled easily by random forest. Also, the performance of the model can be enhanced in case of overfitting with few hyperparameter tuning [25].

2.4.4. Extreme Gradient Boost

According to Prova, Extreme Gradient Boosting is an efficient machine learning model for regression task. Extreme Gradient Boosting is notable for its performance in real-world scale problems using a minimal number of resources since prediction models like decision trees include poor learners rapidly. Previous model addition problems are fixed by training new models. XGBoost with its parameters such as learning rate, subsampling, maximum tree depth and more, the residual errors can be reduced from last iterations. They enhance the generalisability of the model by minimizing overfitting to data with its complex tree building techniques that enables search by split and quick tree trimming. The efficiency of the model and its performance is always important to XGboost. It uses parallel processing, tree pruning, and cache awareness to calculate quicker than gradient boosting implementations [25]. The recent researches in forecasting crude oil prices [9], [27] wind

turbine fault detection framework [28], identifying essential proteins [29], diagnosing chronic kidney disease [30] shows that XGBoost outperforms the other methods achieves better robustness, fast and scalable while presents higher accuracy and a lower false-positive rate. XGBoost can boost prediction performance in terms of specificity, sensitivity, and accuracy. The XGBoost regressor is chosen in this study on students' academic performance due to its excellent performance in machine learning applications[31].

2.5. Model Evaluation

To assess the predictive performance of all the machine learning models, evaluation metrics are used as follows:

1) **Mean absolute error:**

   Mean absolute error (MAE) is defined as the average sum of the absolute differences between the actual value and the predicted value, serving as the measure of average magnitude of errors in a set of predictions, without considering their direction which shows how well a model fits the data [32], [33].

$$MAE = \frac{1}{n}\sum_{i=1}^{n} |Y_i - \hat{Y_i}| \tag{6}$$

   Where n represents the total size of data points,

   Y_i denotes the actual value,

   $\hat{Y_i}$ denotes the predicted outcome [34].

2) **Root Mean Squared Error:**

   Root mean squared error (RMSE) is a commonly used metric for evaluating the accuracy of numerical prediction. It is scale-dependent and unsuitable for comparing prediction errors across multiple variables but it is very helpful for comparing prediction errors of various models for a particular variable. According to [35], RMSE is a metric that measures the discrepancy between the actual values and the predicted values in regression tasks. RMSE is defined as the square root of the average of the all errors squares with following formula,

$$RSME = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(Y_i - \hat{Y}_i\right)^2} \quad (7)$$

Where n signifies number of data points,

Y_i represents the actual value,

$\hat{Y}_i$ represents the predicted outcome [34].

3) R squared:

R squared is a statistical measure for evaluating the goodness of fit in regression models. "It measures the proportion of the total variation in the dependent variable that is predictable from the independent variable". R^2 can be mathematically defined as [36]:

$$R^2 = 1 - \left(\frac{SSR}{SST}\right) \quad (8)$$

Where SSR is Sum of Squared in which represents the difference between the actual data points and the predicted values captured from the regression model.

SST is Total Sum of Squares, which measures the total variation of the actual values from the overall mean value of the dependent variable.

The coefficient of determination R2 ranges from 0 to 1:
If $R^2 = 0$, the model explains none of the variability;
If $R^2 = 1$, the model explains the all of the variation.

2.6. Feature importance analysis using SHAP

This research paper uses explainable artificial intelligence analysis to strengthen the interpretability of the machine learning framework [37], [38]. The key factors impacting the educational success of students will be predicted by applying feature importance

analysis to machine learning algorithms. These models will be incorporated with SHapley Additive Explanations (SHAP). SHAP analyses and explain the overall impact of input features on model outcome as well as the individual contribution of each feature especially presents the order of feature. This eliminates the existing black box system and ensures transparency in decision making. This achieves meaningful insights for educators. SHAP solves the computing task of prediction contribution of a model for every subgroup in the dataset [8], [39]. According to recent research, implementation of SHAP values with XGBoost regression training has explained the functionality of the model for a single target data [39]. SHAP can provide both local and global interpretability offering fair feature attribution across datasets. Moreover, SHAP offers consistency and accuracy by providing unique solutions that satisfies mathematical properties.

2.7. Visualization and interpretation

Recent advancement in learning involves learning Analytics Dashboard (LAD). The educators and students often provided with LAD for uncovering the patterns of learning behaviour of students in online platform. In order to effectively analyse the learning process and help students to achieve their learning objectives, Learning Analytics (LA) provide a view for identifying learning trends and patterns from an increasing volume of student information [35]. Finding patterns that are not evident with numerical representation are made simple and easy to understand using visual representation which include using various chart types. Visual data analytics is a method for evaluating and visualizing data which focus on business intelligence (BI). Few popular tools include Tableau, QlikView, Google Data Studio and Power BI. The tools use visualizations using bubble chart, radar chart, heat maps, Sankey diagram and maps with open-source libraries like Matplotlib, Pyplot and Seaborn. The main purpose of the research paper is to provide insight about student academic success using simple as well as complex visual chats [40]. Also, using BI tools to create interactive dashboard will improve user engagement by revealing hidden learning patterns.

Chapter 3 – Results

3.1. Data Preparation results

While preparing the dataset with the initial checks, it is found that there are no missing values. An empty unnamed column with no observations was removed.

3.2. EDA results

3.2.1. Outlier detection

Interquartile Range (IQR) method was used for Outlier detection among the numerical variables. The boxplot is used to represent the outliers. From the fig. 3, fig. 4 and fig. 5 outliers were observed in the academic scores including both secondary examinations as well as the university competency scores - ‘MAT_S11', 'CR_S11', 'CC_S11', 'BIO_S11', 'ENG_S11', 'QR_PRO', 'CR_PRO', 'CC_PRO', 'ENG_PRO', 'WC_PRO' and the global score 'G_SC'.

From the boxplot, the observed outliers (beyond the whiskers) represent the extreme performers. The median scores of each subject show differences in student performance distributions. The target variable G_SC (Global Score) contains less outliers.

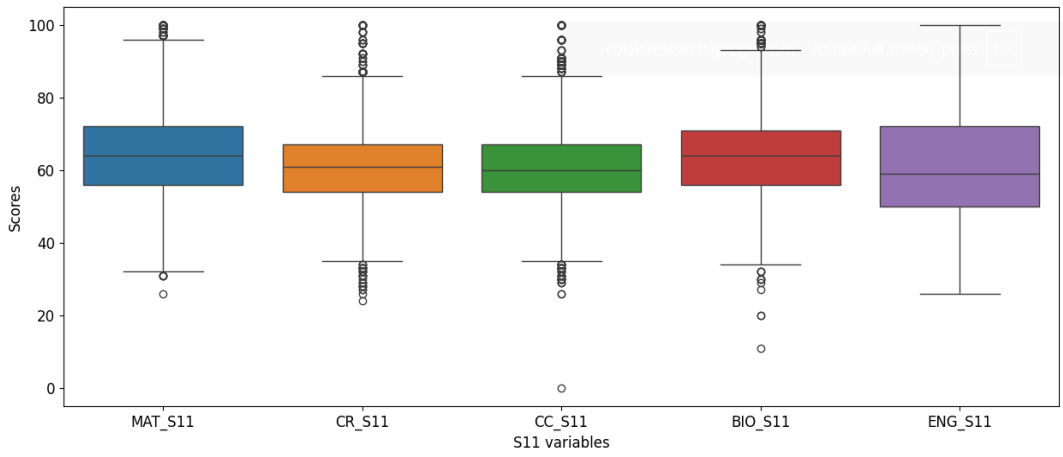


Figure 3. Boxplot for Saber11 variables.

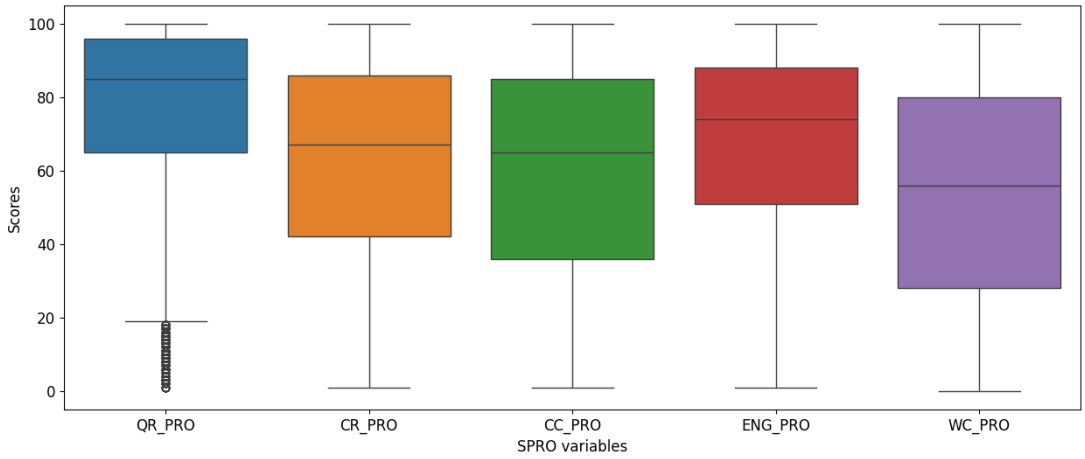


Figure 4. Boxplot for SaberPRO variables.

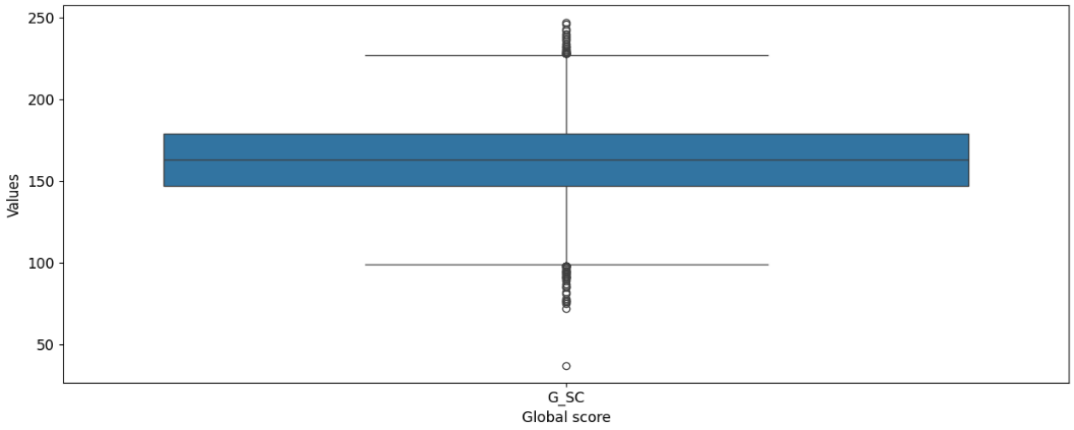


Figure 5. Boxplot of global score.

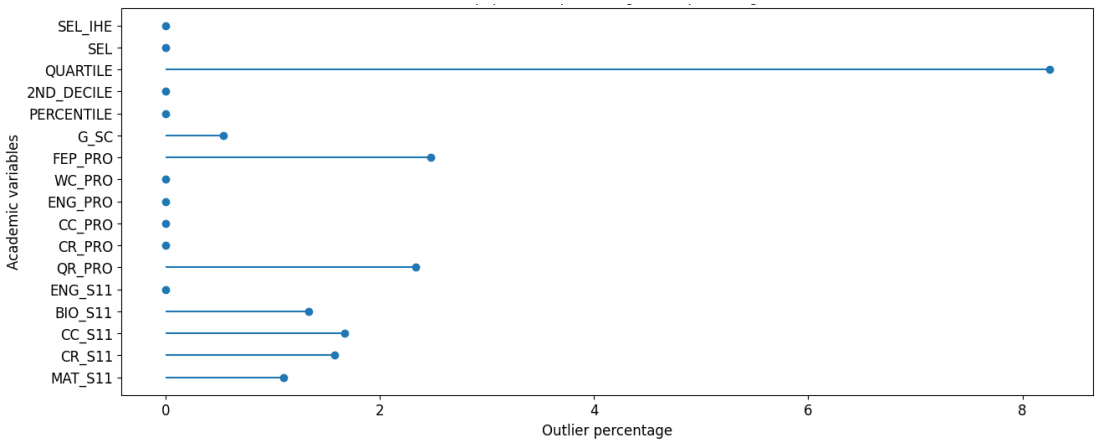


Figure 6. Outlier percentage analysis.

The outlier percentage analysis is represented using Lollipop chart. The outliers detected using IQR method is quantified in terms of percentage. From the fig. 6, it reveals that the ration of outliers is less than 3% except QUARTILE. These observation from the boxplot

and the lollipop chart represent high and low performing students rather than data quality issues. Thus, outlier removal is not performed ensuring data quality for machine learning modelling.

3.2.2. Correlation matrix

The fig. 7 is the correlation heatmap which is used to analyse the linear relationship among the numerical variables before conducting regression or predictive modelling. The correlation matrix revealed that the degree of association between variables are positively related with no negative or week relationships. The higher secondary subjects range from r=0.55 to r=0.76 exhibiting moderate to strong positive correlations. Among them, Mathematics and Biology have the strongest relationship (r=0.76) while creative writing and communication has 0.73. The S_11 scores and the S_PRO scores are moderately correlated with each other. It is clear that the target variable G_SC global score exhibits moderate to strong positive relationship with each academic variable. The strongest correlation among the academic ranking variables occurs among G_SC, Percentile, Second Decile and Quartile. In comparison with other variables, WC_PRO and FEP_PRO shows weak relationship.

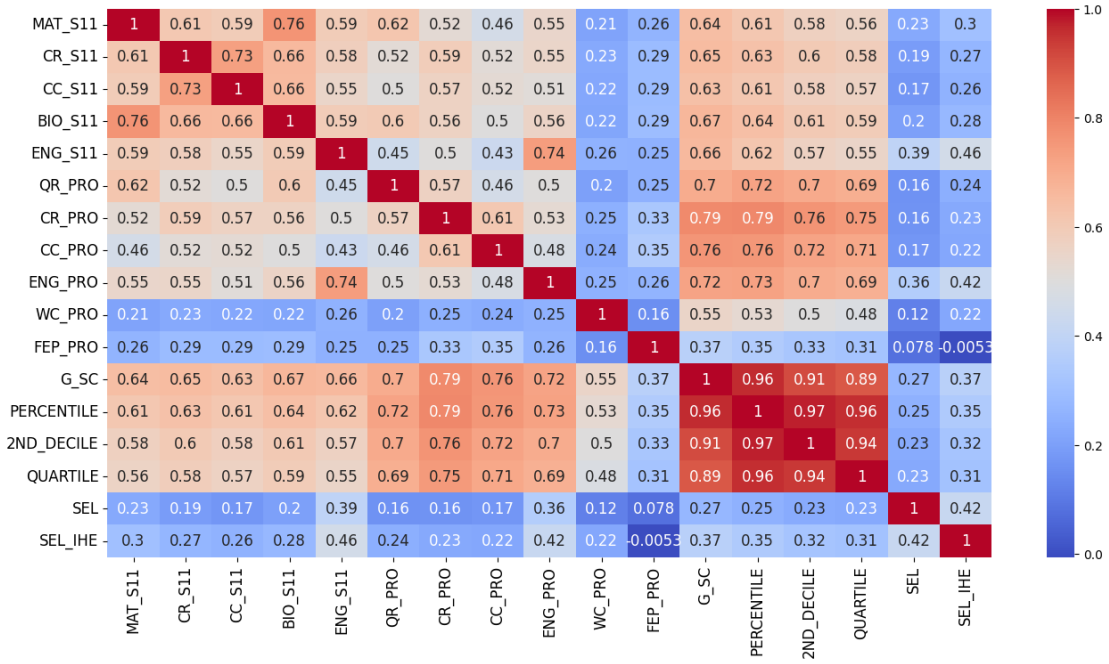
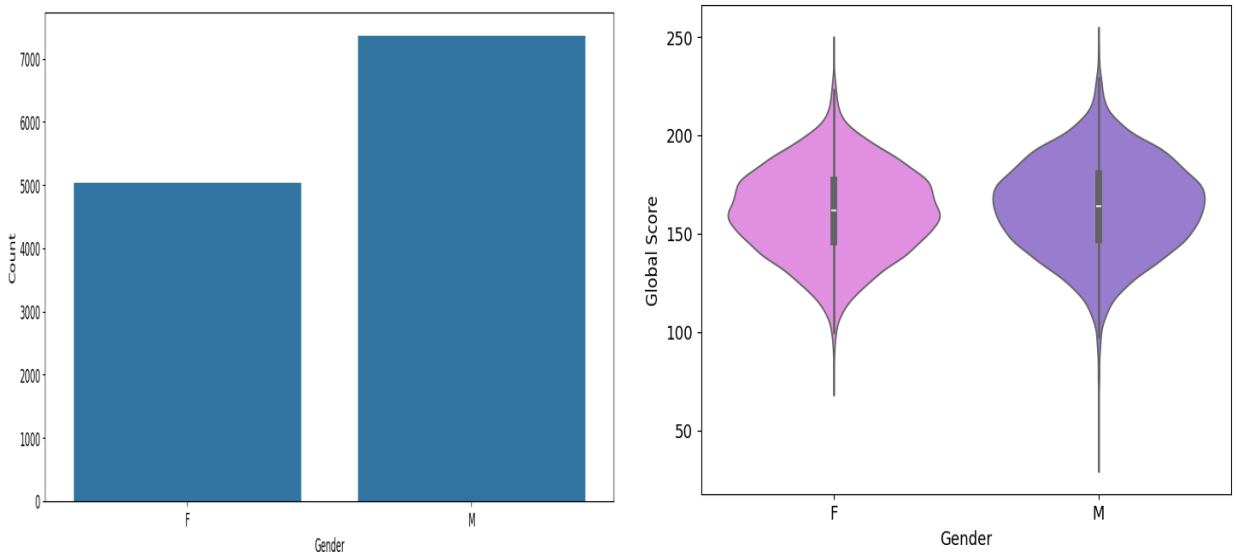


Figure 7 Correlation heatmap of numerical variables.

3.2.3. Distribution of global score across gender

The fig. 8. (a) demonstrates the students across gender. Though there is a class imbalance, the goal is to predict a continuous numerical value. Thus, this imbalance is ignored. The fig. 8. (b) represents the distribution of global score between male and female students using violin chart with embedded interquartile range. Both male and female students exhibit almost similar score distribution. The density of observation is concentrated between 150 and 180 while the median scores are approximately between 160 and 165 for both the genders. The narrow endings reveal the low performers slightly more in male students. Since the observations at the extremes are limited, there is no gender influence on the academic performance of students.



(a) (b)

Figure 8. Student demographic and performance distribution.
(a) Distribution of gender. (b) Distribution of global score by gender.

3.2.4. Analysis of academic factors

The analysis of academic factors starts with the university. The fig. 9 illustrates the top 10 universities with average global score of students.

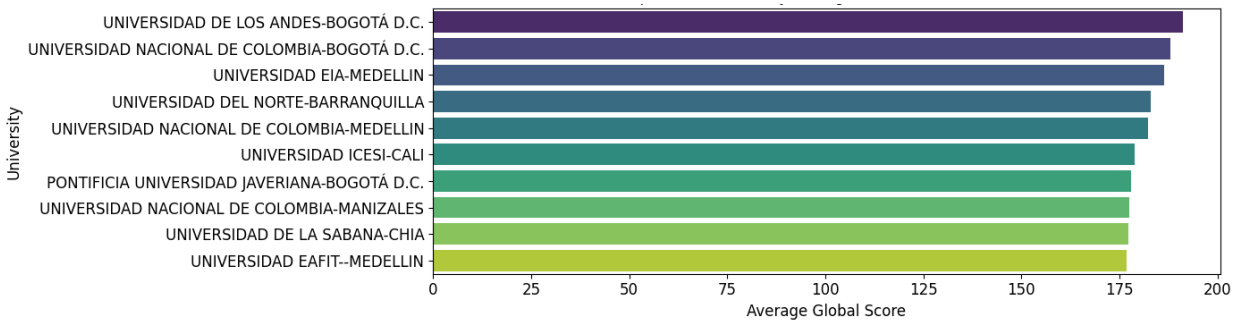
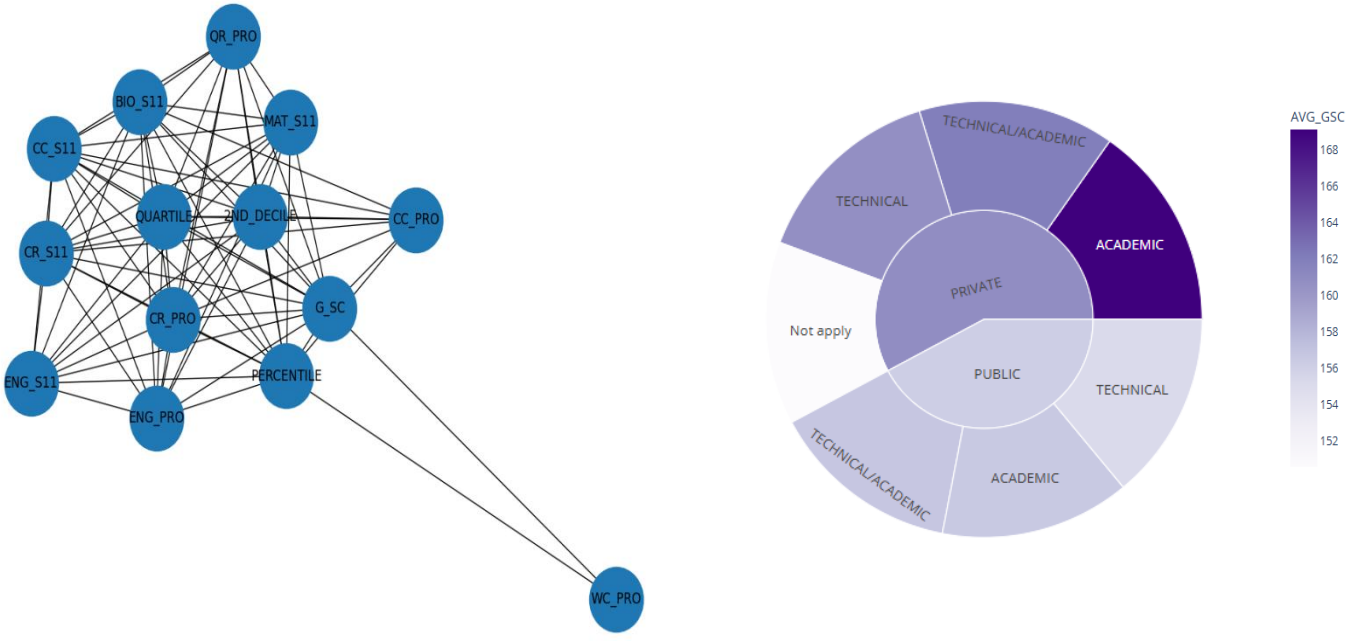


Figure 9. Top 10 universities by academic performance.

The network graph in the fig. 10. (a) is used to display the interdependence of academic variables and to show how strongly they are related is shown in the figure. From the visual positions, the academic variables CC_S11, MAT_S11, CR_S11 and G_SC are identified as the key variables with strong association while WC_PRO falls in an isolated position indicating less strongly associated variable for the academic performance. Though the network provides the understanding of the variables it does not provide ethe magnitude of the correlation coefficients.



(a) (b)

Figure 10. Top 10 universities by academic performance

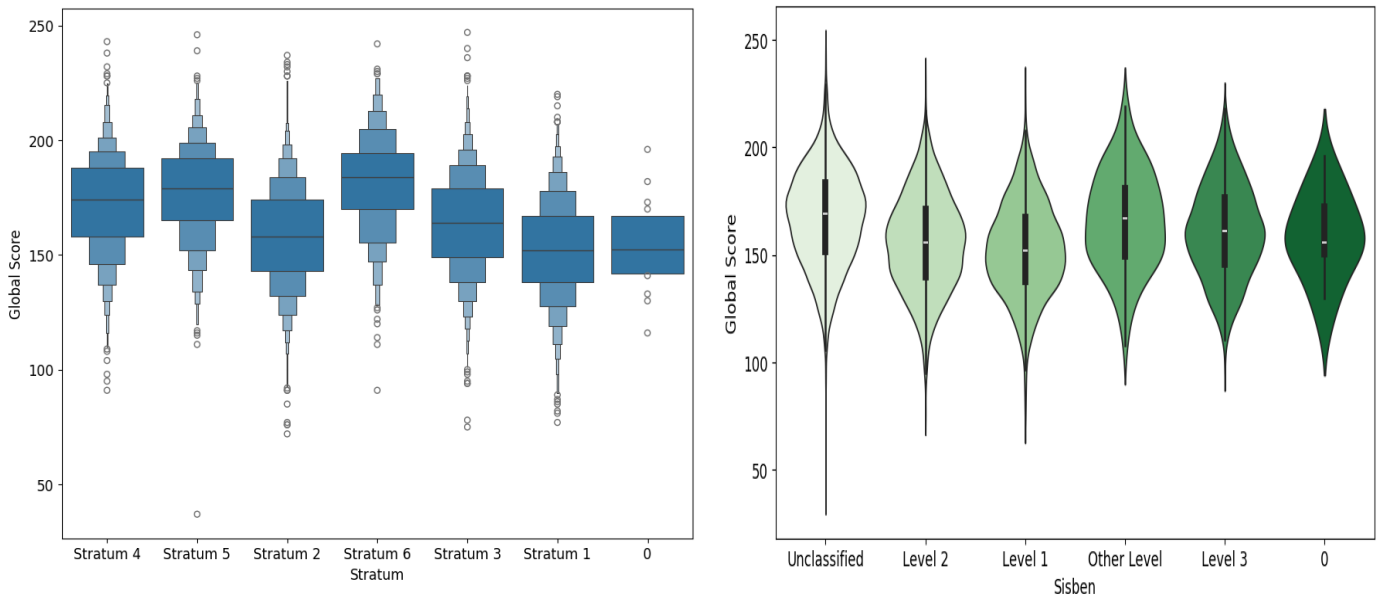
(a) Correlation network of academic scores

(b) Sunburst chart representing the impact of school nature and type.

The fig. 10. (b) is a Sunburst chart representing the impact of school nature and type where private school dominates the public school.

3.2.5. Distribution of academic performance by socio economic factors

The fig. 11. (a) uses the boxen chart to illustrate how the stratum influences the academic performance of students. The visual reveals from the central tendency that the students from higher strata performed academics with higher scores than the lower strata. The median scores of students from Stratum 1 and Stratum 2 falls between 140 to 165 while the students from stratum 3 scores relatively higher occupying an intermediate position between higher and lower strata. Stratum 4 and Stratum 5 illustrate students with higher median scores concentrated between 160 and 185 than the lower categories. The student from stratum 6 displays the highest median scores ranging from 170 to 195. There are outliers in almost every category above 220 displaying the exceptional performers from different backgrounds. The stratum 5 has very few students with the very low academic performance scoring below 50.



(a) (b)

Figure 11. Distribution of academic performance by socio economic factors.

(a) Distribution of academic performance by Stratum. (b) Distribution of academic performance among SISBEN levels.

The fig. 11. (b) is a violin plot illustrating how the SISBEN influences the academic performance of students. The visual reveals from the central tendency that medians are relatively similar across groups. The groups – Unclassified and other level demonstrate broader distribution when compared with other levels. The substantial overlapping of scores across different groups says that it is not a strong predictor of Global Score.

The fig. 12 display the distribution of academic performance of students from different revenue backgrounds. The pie chart illustrates that the distribution of all income groups is balanced with proportions ranging from 11.5 % to 13.7 %. The highest contributor is the students from revenue background where Legal Monthly Minimum Wage is 10 or more whereas the lowest contributor belongs to the category with less than 1 LMMW.

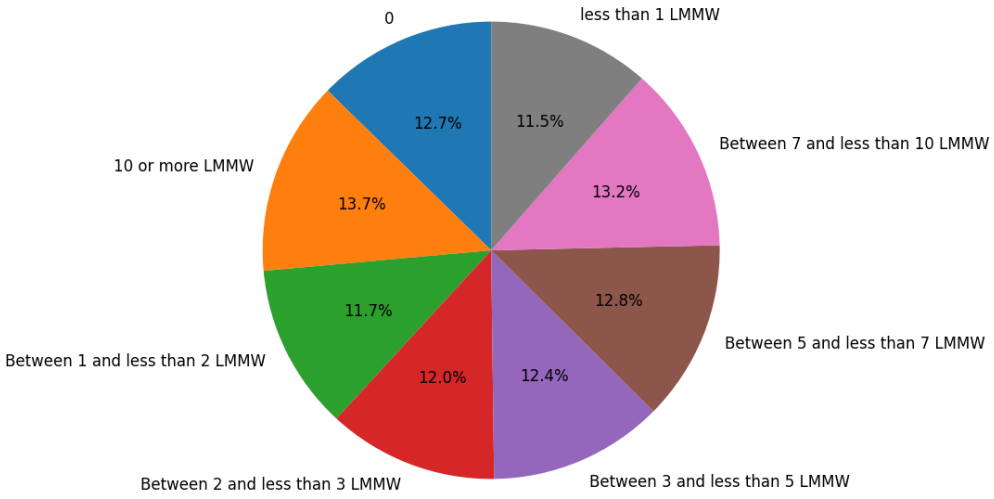


Figure 12. Impact percentage of revenue on academic performance.

The radar chart shown in the fig. 13 represents the comparison of high performers and low performers based on the socioeconomic factors - parents education, occupation and the household resources availability. This visual reveals that the high performers achieve high scores than the at-risk students across all the factors. There is a strong influence on the students with household resources and parents with educational qualifications. Although

the parent occupation displays a positive relationship with the academic score, it possesses weaker influence compared to the other two socioeconomic status.

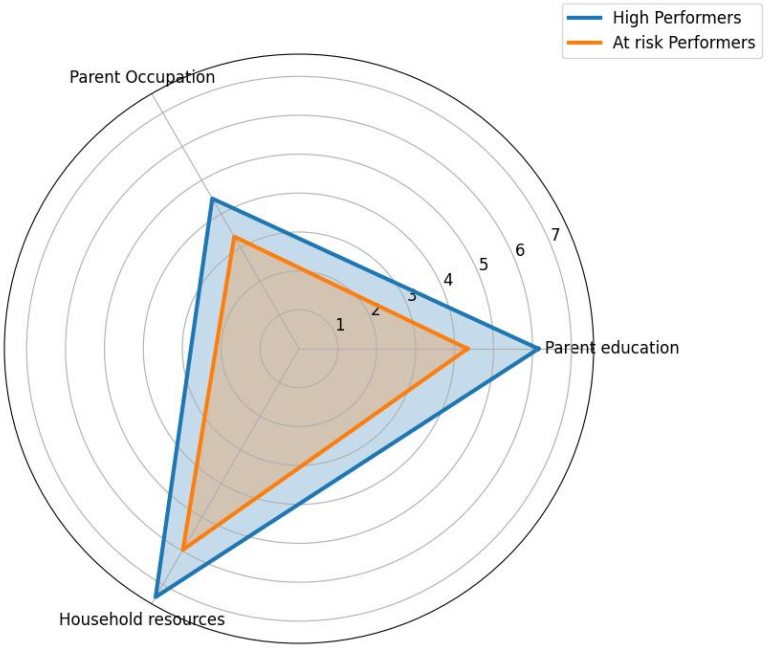


Figure 13. Socioeconomic impact on academic result.

3.2.6. Distribution of academic performance by household resources

Household resources is the feature engineered variable with variables including 'TV', 'CAR', 'COMPUTER', 'DVD', 'MIC_OVEN', 'PHONE', 'MOBILE', 'WASHING_MCH' and 'INTERNET'. The relationship between the student's global score and the household resources is illustrated in the fig. 14 using bubble chart. A clear pattern can be observed from the visual that the size of the bubble increases gradually with higher household resource score. The students with lowest household resources achieved scores between 152 and 155 while the students with highest resource score attained scores ranging from 164 to 172. This visual strongly displays the strong and clear positive relationship between household resources and academic performance.

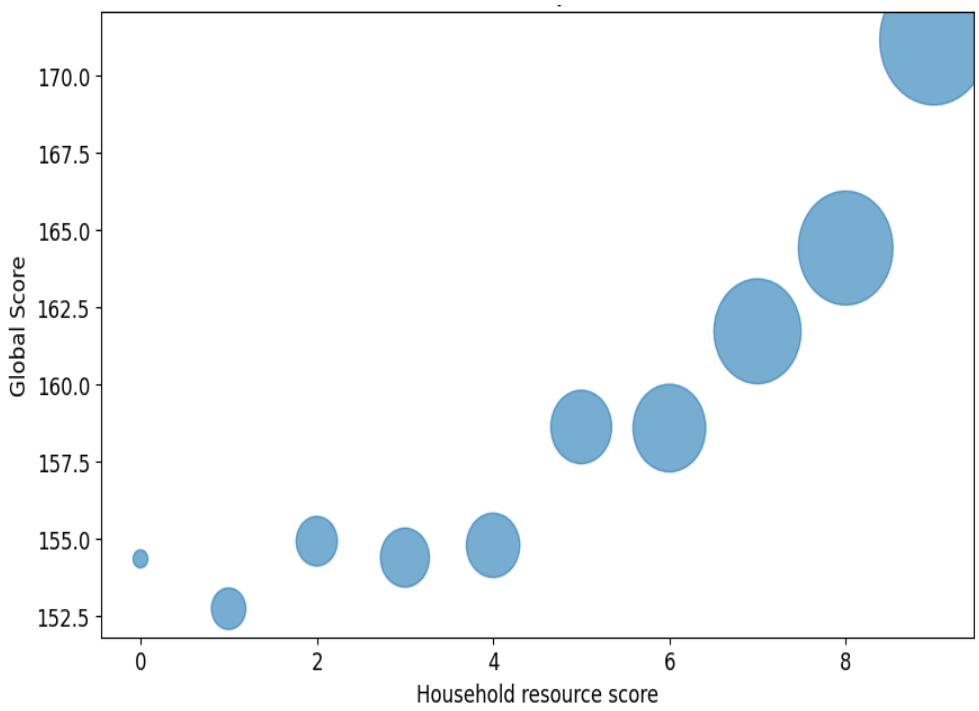


Figure 14. Global score distribution by household resources.

3.3. Encoding

Categorical variables such as binary, nominal, or ordinal were encoded with encoding methods before machine-learning modeling. The binary variables in the dataset are GENDER, INTERNET, TV, COMPUTER, WASHING_MCH, MIC_OVEN, CAR, DVD, PHONE, MOBILE, SCHOOL_NAT are encoded into numerical format that fits the predictive model.

The variables such as SCHOOL_TYPE and ACADEMIC_PROGRAM were encoded using using one hot encoding method since each had more nominal categories. Instead of representing them with irrelevant ranking, a distinct binary value was used to represent each category. The original variables were dropped from the data frame to minimize data redundancy, especially to prevent multicollinearity in linear regression model.

The ordinal encoding method was applied to variables such as EDU_FATHER, EDU_MOTHER, OCC_FATHER, OCC_MOTHER, STRATUM, SISBEN, PEOPLE_HOUSE, REVENUE and JOB. The assignment of numerical values was

depicted in increasing order of their category instead of irrelevant nominal representation. This allows meaningful ranked information for modeling the dataset.

3.4. Feature Engineering and Model Preparation

The main purpose of feature engineering was to capture trends that are not sufficient in the dataset. S11_AVG is the average of SABER 11 subjects which is featured to determine to assess the past academic achievement at school level. Similarly, PRO_AVG is the average of the professional scores. Moreover, ACADEMIC_GROWTH is the engineered variable by computing the difference between PRO_AVG and S11_AVG capturing the performance changes between two stages.

The parental features such as PARENT_EDU_INDEX, PARENT_EDU_GAP AND PARENT_OCC_INDEX were created from the independent variables of mother and father to reflect parental educational and occupational background. The computation of combined parental features gives meaningful addition to the model rather than individual background values.

The individual variables such as INTERNET, TV, COMPUTER, WASHING_MCH, MIC_OVEN, CAR, DVD, PHONE, MOBILE contributing to the household resource availability were combined to derive a composite variable called RESOURCE_SCORE. The final score offers a picture in socioeconomic status.

In order to make sure that the prediction models do not encounter target data leakage, an evaluation was done to check the prediction models were not using variables that are directly linked with the target variable. With the six SABER PRO variables, the evaluation demonstrated R2 of 0.946 approximately which showed strong relationship and multicollinearity with the target variable G_SC. To overcome this scenario, SABER_PRO modules along with engineered features such as ‘PRO_AVG’ and ‘ACADEMIC_GROWTH’ were removed from the final data frame for predictive modeling.

3.5. Predictive modelling results

Predictive models used to predict the global score were supervised regression models: Linear Regression, Decision Tree, Random Forest and XGBoost. The train data and test data was divided using 80:20 split with random state 42. The 4 models were trained and the training and testing performance of each model was evaluated to assess potential overfitting or underfitting with three metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and R squared (R2).

TABLE III. COMPARISON OF MODEL PERFORMANCE.

Model	MAE (Train)	MAE (Test)	RMSE (train)	RMSE (test)	R2 (train)	R2 (test)	R2 Gap
Linear regression	10.98	10.94	14.27	14.31	0.62	0.61	0.01
XGBoost	10.72	10.90	13.95	14.31	0.64	0.61	0.03
Decision tree	10.87	11.12	14.19	14.56	0.62	0.59	0.03
Random forest	10.57	10.93	13.80	14.36	0.65	0.61	0.04

From the above table, the comparison results show that linear Regression model achieved the best performance with R2 of 0.62 and 0.61 on training and testing data respectively. Stable generalisation was observed with very small R2 gap of 0.01. The marginal changes between training and testing sets of RMSE and MAE from 14.27 to 14.31 and 10.98 to 10.94 indicated consistent prediction errors.

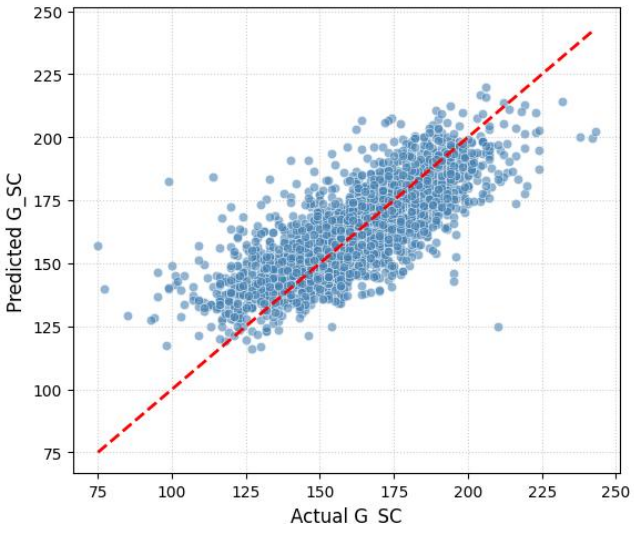
Comparatively, XGBoost reached second best overall performance with closer results to linear regression. With R2 gap of 0.03 where R2 score with train and test data were 0.64

and 0.61 the model fitted better. The test MAE of 10.72 and test MAE of 14.31 stayed closer to linear regression results which showed better generalizability.
The Decision Tree model produced the weakest predictive accuracy of all the other models with highest test MAE and RMSE of 11.12 and 14.56 respectively. Though the R2 gap is small, this model is comparatively poor.

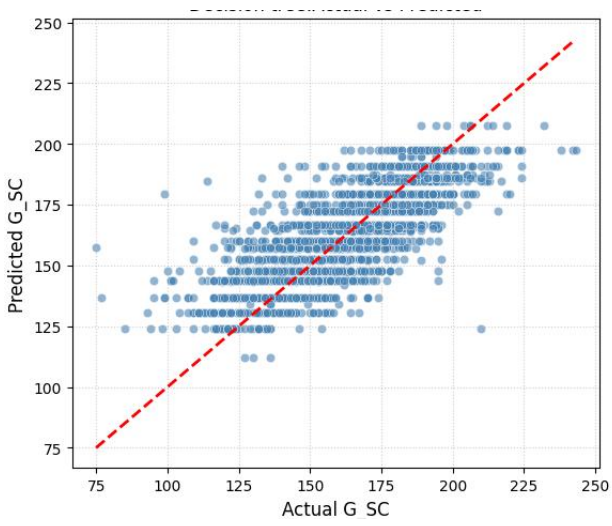
Though the test MAE of 10.93 and RMSE 0f 14.36 of Random Forest model remained competitive, the model achieved highest train test set gap of 0.o4 with train R2 of 0.65 and test R2 of 0.61. This indicates the model has greater fitting to train data but does not cause severe overfitting.

Consequently, the findings show that the models produced moderate predictive performance with respectable generalisation rather than overfitting or underfitting. Also, it is evident that the complexity of the model did not necessarily improve prediction. The linear regression as best performer shows significant portion of the relationship between the influencers and target global score were represented with linear model.

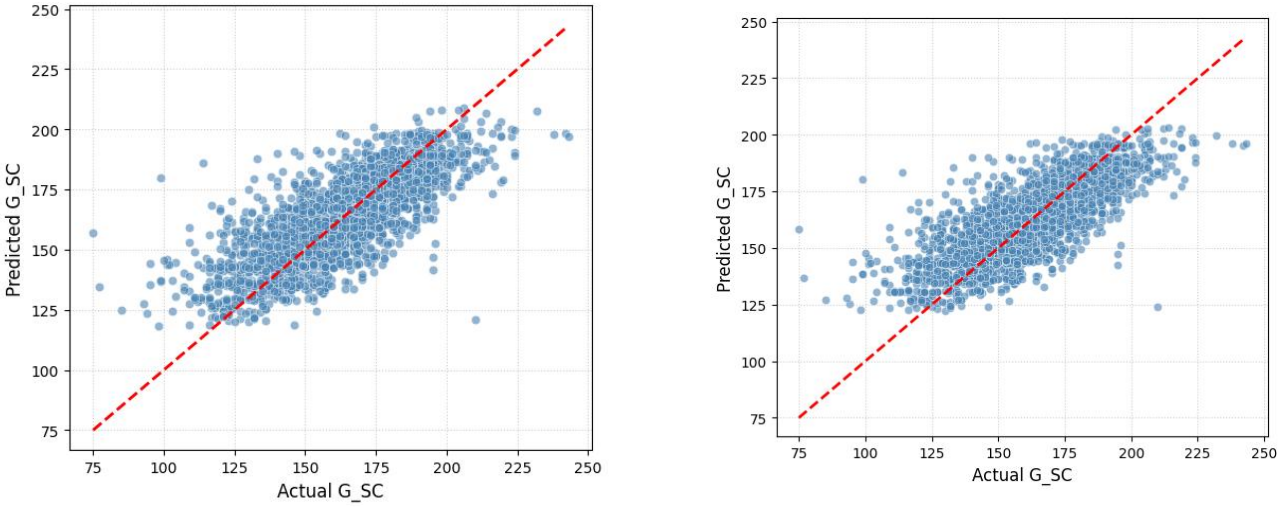
The actual values and predicted values of Global scores from all four models were plotted in the fig. 15. (a), (b), (c) and (d). The diagonal red line is the reference line whereas observation points closer to this line are the predictions with less errors. The visual observation around the reference line indicates the model's ability to capture the variance in academic performance while the deviations are the prediction errors.



(a)



(b)



(c) (d)

Figure 15. Actual G_SC Vs Predicted G_SC.

(a) Linear Regression (b) Decision Tree

(c) Random Forest (d) XGBoost

3.6. Feature Importance Analysis

The analysis of feature importance was performed for all four models to highlight the important predictors influencing the student academic performance. The feature importance analysis in linear regression revealed that S11_AVG was an important academic factor. Predictions were also influenced by parental socioeconomic factors such as PARENT_EDU_INDEX, PARENT_OCC_INDEX. RESOURCE_SCORE is another important factor revealing the prediction. The direction of relationship was revealed directly by the coefficients.

When it comes to decision tree, the parental education, occupation and household resources contributed to the predictions, academic factors dominated the tree splits. The results of feature analysis on Random Forest showed that the academic factors remained significant even though the socioeconomic features were providing additional predictive information. The XGBoost feature analysis had socioeconomic features and S11_AVG as the significant predictors. Further evidence of how these features influenced individual predictions and their direction of relationship were demonstrated in SHAP analysis.

Overall, the findings revealed that the prior academic performance and socioeconomic factors were beneficial for predicting the student academic performance. However, these results showcase the predictive causation instead of causation.

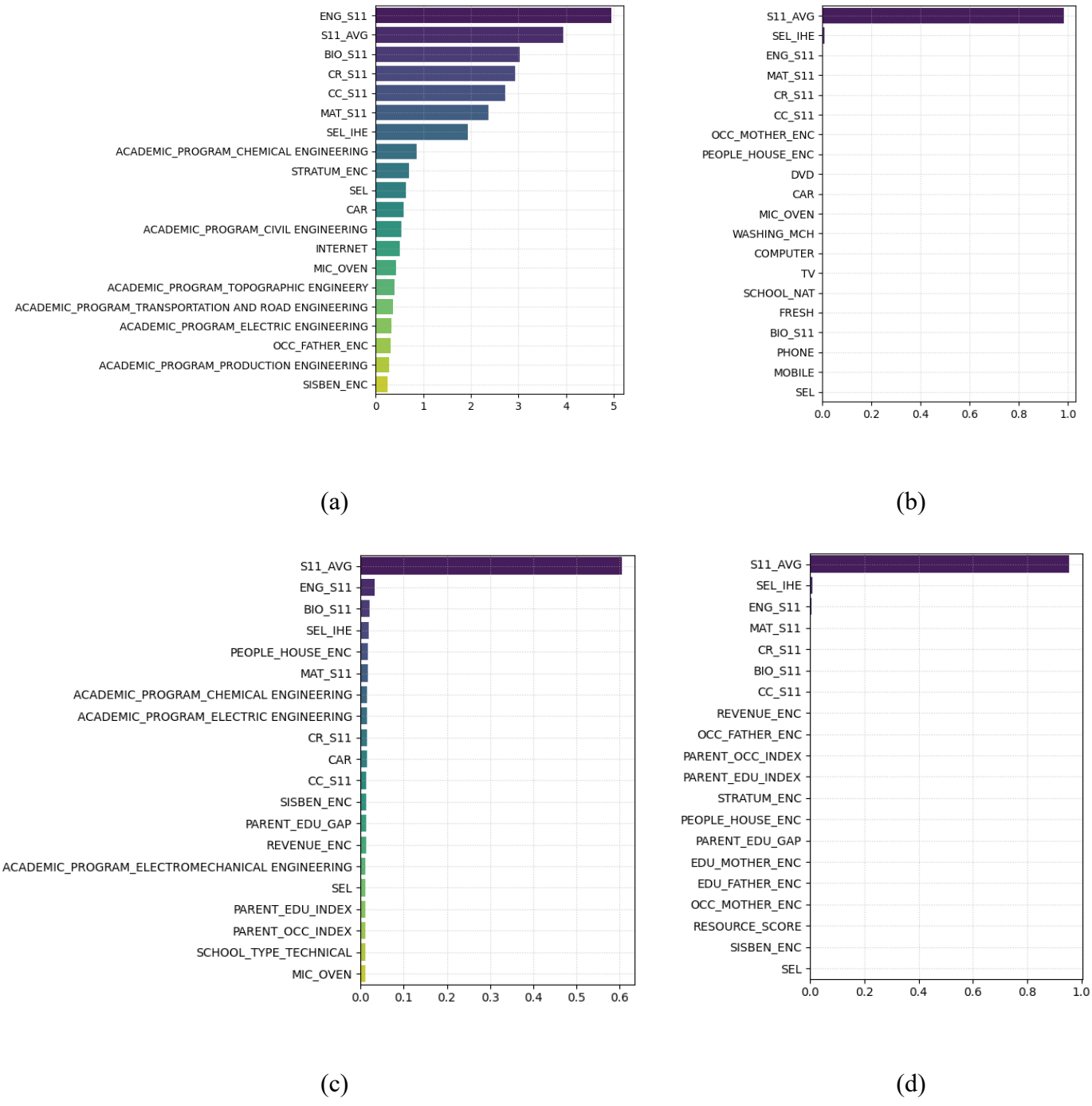


Figure 16. Feature importance analysis.

(a) Linear Regression

(b) Decision Tree

(c) XGBoost

(d) Random Forest

3.7. SHAP results

The XGBoost model was subjected to SHAP analysis to improve the interpretability and identify the most significantly influencing features of global score. The results revealed that the prior educational achievement S11_AVG with ENG_S11, BIO_S11, CR_S11, CC_S11, and MAT_S11 were the most influencing factors. Although the socioeconomic and environmental factors like SEL_IHE, SEL, INTERNET, MIC_OVEN and SISBEN were of lesser degree, they also influenced the model.

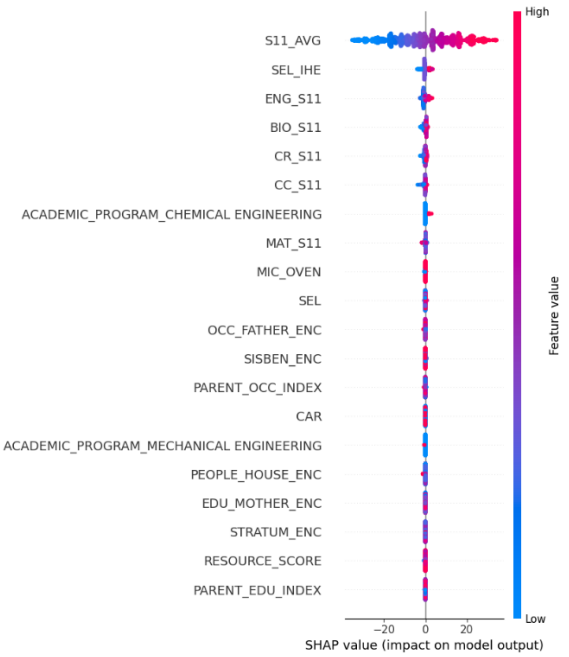


Figure 17. SHAP summary and bar plots.

The SHAP summary plots rank the features based on their mean absolute contribution to the predictions of the model. The positive and the negative SHAP values indicate the individual observations drive the predictions above or below the models predicted value of global score while the higher ranked features have greater overall impact. The predominance of SABER 11 educational factors indicated that the model XGBoost produced primary predictive signal as previous academic achievement.

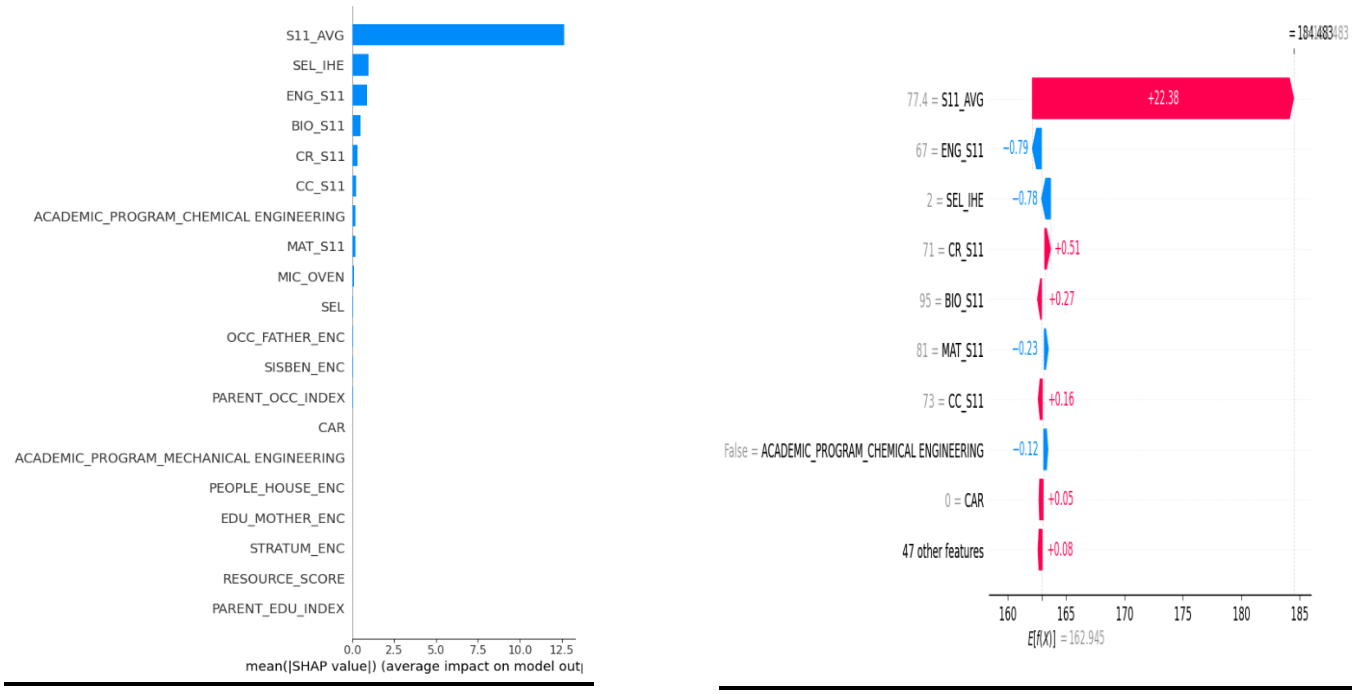


Figure 18. SHAP plots.

(a) SHAP bar plot

(b) SHAP waterfall plot.

The following fig. demonstrates the SHAP dependence analysis. The relationship between the changes in the influential factor and the changes in its contribution to the global score is depicted in the SHAP dependence analysis. Moreover, a local SHAP explanation suggests how individual feature values impact a particular prediction in relation to the baseline of the model. This analysis provided transparency to model predictions as well as to the characteristics of the model.

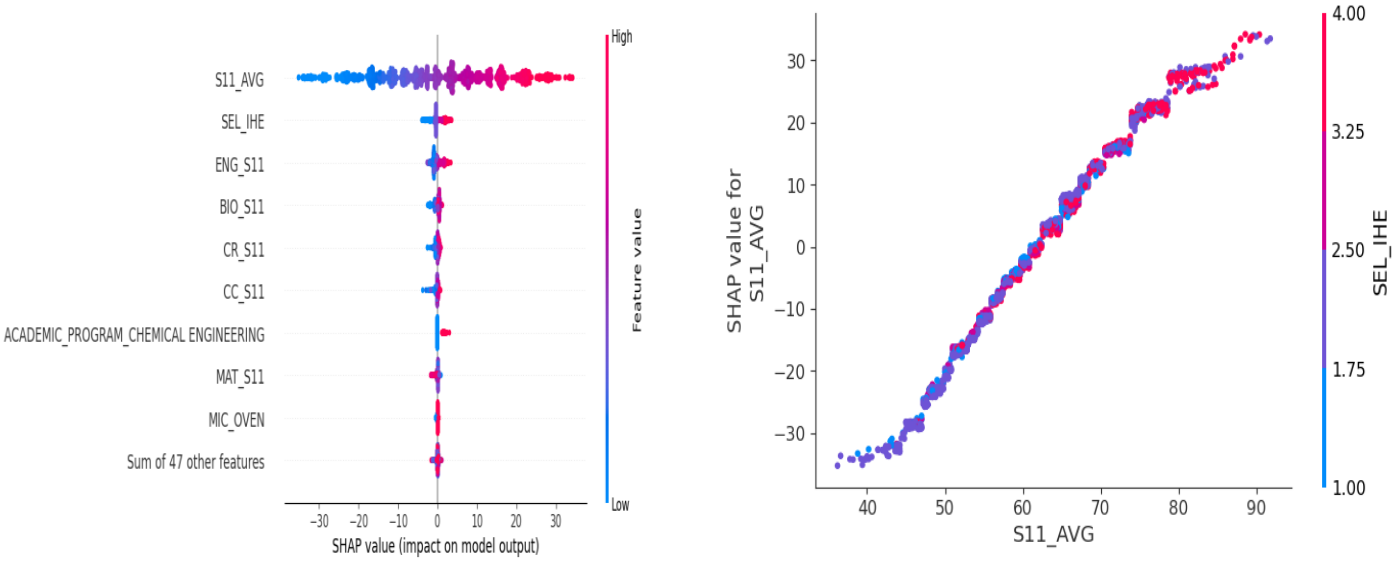


Figure 19. SHAP dependence and beeswarm plots.

(a) SHAP beeswarm plot

(b) SHAP dependence plot.

Overall, the SHAP analysis revealed that previous academic performance was the most significant factor influencing the prediction, with additional factors from environmental and socioeconomic status. However, the results should not be considered as a proof that changing specific characteristic will directly result in Global score change. This is because SHAP values denote model-based associations rather than casual effects.

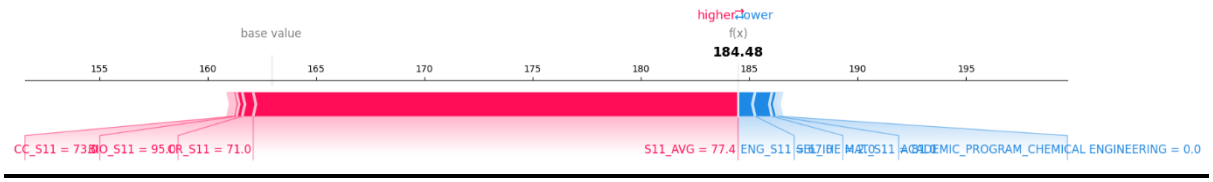


Figure 20. Force plot.

3.8. Dashboard results

The dashboard is created for visual analytics that provides an integrated vision of student's patterns, academic growth and predictive modelling. The dashboard hols three pages namely - "Student Landscape", "Academic Journey" and "ML Prediction and Explainability".

Firstly, the Student landscape dashboard displays the average global score of 162.71 with significant variations across different socioeconomic groups. The students with previous academic achievement S11_SVG exhibit a positive relationship to the targeted global score. Average performance increases from **152.35** in the lowest socioeconomic stratum to **181.51** in the highest. Students from lowest socioeconomic class, exhibit a rise in performance from 152.35 to 181.51.

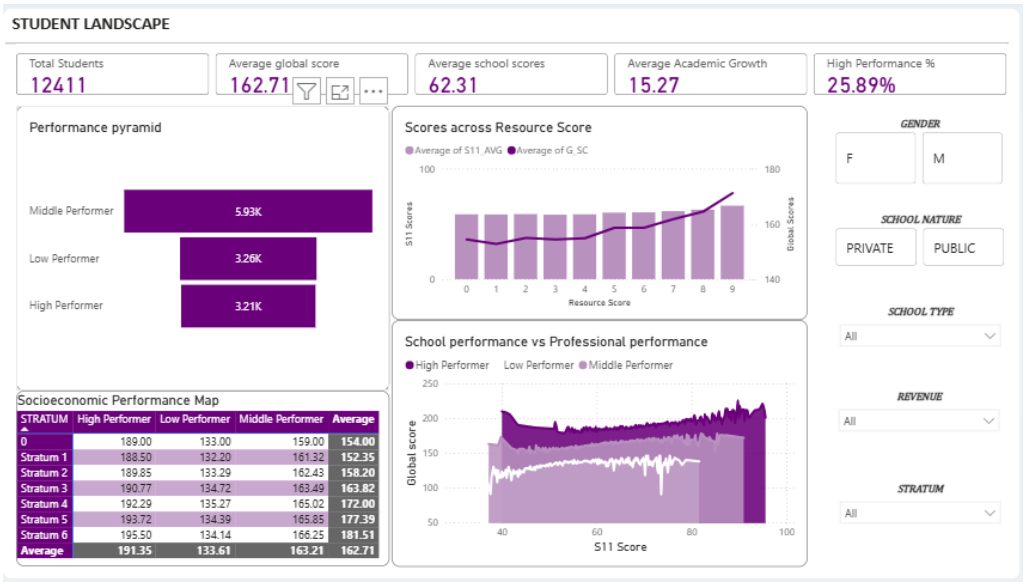


Figure 21. Student landscape.

The second page "Academic Journey" visualises how the academic journey of students depends on environments like job status, revenue category and parental educational background. The students from household income category of ten or more LMMW scores an average of 183.67 which is a rise from 153.31. The students from parental educational backgrounds with higher index tend to achieve better results. The academic schools score higher average score of 165.85 than technical schools with 156.86.

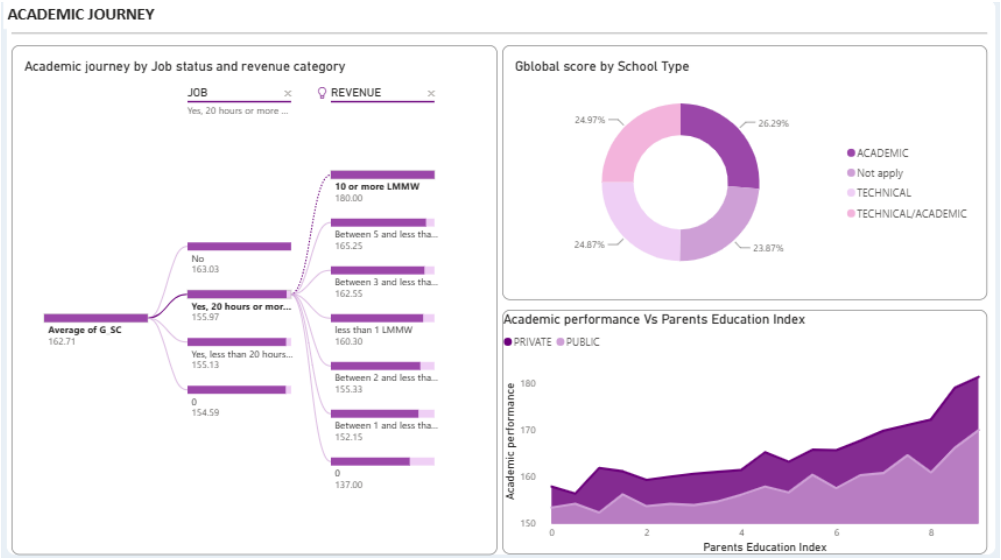


Figure 22. Academic Journey.

The final visual dashboard depicts the ML prediction with explainability. With R squared value ranging between 0.59 and 0.61, the performance of the predictive models was moderate. The linear regression model yields the best performance with R squared of 0.61 while XGBoost produced second best model with MAE of 10.89. The prior academic performance of SABER 11 outperforms other features with mean absolute SHAP value 12.66 in the SHAP analysis.

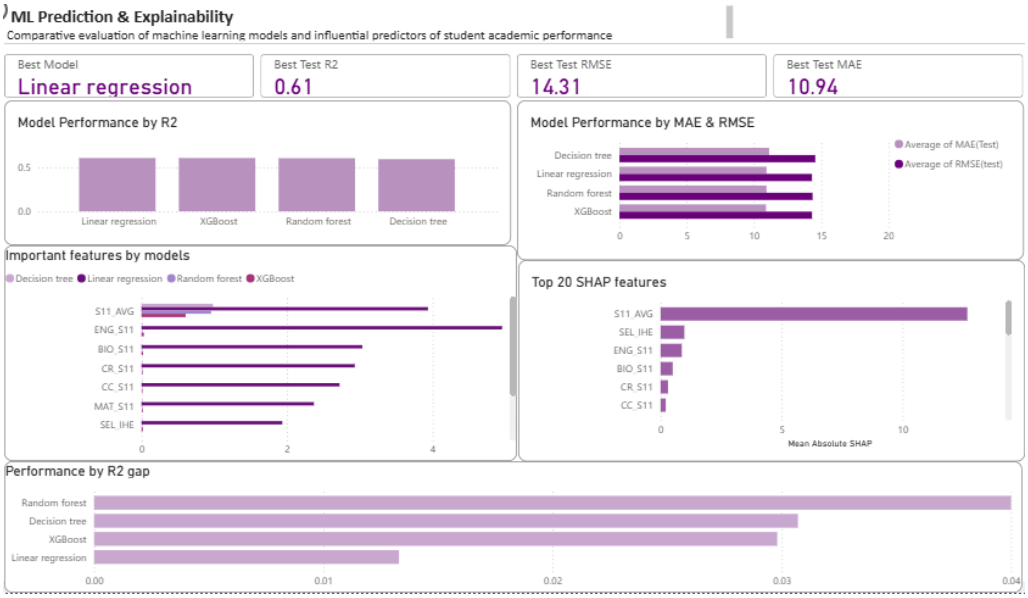


Figure 23. ML Prediction and explainability.

Overall, the educators can be beneficial from the dashboard to analyse the significant influence of socioeconomic environments, household income, resources, school type and nature, and parental backgrounds on the student performance. Although the dashboard suggests the past academic achievement as the highly reliable predictor of performance.

Chapter 4 – Discussion

The objective of this study was to examine if the combination of regression models with feature importance analysis, SHAP and dashboard analytics could provide precise and interpretable results of student academic performance prediction. Among the four models used, Linear Regression performed the best with highest R-squared value of 0.61 marginally higher than XGBoost (MAE 10.98). The yielded XGBoost results closer to linear regression indicates that results were against the assumption that complex models outperform linear models. The models produced moderate results with R squared value ranging from 0.59 to 0.61 with almost 40 percent of variation in the Global Score which remains by the existing features. This gap in the model indicates missing of elements like mental well-being, attendance, study habits, etc. This study showed that explainability techniques with machine learning techniques can yield significant insights into academic performance of students. The feature importance analysis and all four models and SHAP were carried out separately but the results were found to be almost same. S11_AVG which is the consolidated previous academic achievement of students found to be the most significant factor of Global core, underscoring the importance of past academic success in predicting future results. The socioeconomic features like parental education, household revenue status, and resources availability were similarly associated to academic achievement, reflecting the dashboard with average scores of 152.35 to 181.51 across different strata. Though socioeconomic features exhibited less predictive power, they were consistently establishing secondary relationship with global score. By displaying the student profiles along with the socioeconomic trends and model results with explainability metrics in an interactive way, the dashboard overcomes the educator's struggle. This handles the problem in the research by yielding the accurate results of predictive models and also give educational stakeholders explanations. It is also important to notice that the results of the feature importance and SHAP provides the relationship of the factors that the model depends to predict the Global score but that does not explain the causation if the factors were changed. The use of target data leakage aware framework acted as the key strength to this project. The framework checked and removed the SABER PRO features that are directly associated with the target variable global score, enhanced the dependability of model evaluation and avoided multicollinearity and artificial model. The implementation of dashboard using Power BI tool empowers the educators with much clear understanding from different aspects.

Ethical considerations

The use of dataset with personal and socioeconomic data of students for predicting their academic performance raises ethical issues that are addressed below.

- **Data source:** The dataset used for this examination is a free to use, publicly available research dataset from Mendeley Data Repository which is an academic data platform. The Colombian Institute for the Evaluation of Education (ICFES) collected and adjusted the information of this dataset with SABER 11 and SABER PRO assessments. The original dataset is cited in the References.

- **Data privacy:** The data provider has de-identified the student information with academic as well as socioeconomic of 12,411 students prior to publication. The personally identifiable information of students like name, address or any other details was not collected and accessed. Also, the socioeconomic factors in this dataset were available as categorical band. Therefore, the analysis is considered to be done on anonymous data.

- **Model bias and fairness:** The consideration of socioeconomic features as secondary predictive weight might favour advantaged students if the model is used properly. The target data leakage pipeline and SHAP explainability provides the prediction transparent and reveals the relationship.

- **Methodological integrity:** The methodology followed in this study from data collection to modelling and analysis were built with no tutorial or external codebase. The target leakage features were identified independently with the correlation and the evaluation metric R-squared. All these methodologies implemented were original with no reference sources.

- **Reproducibility:** The code notebook provides inline explanations for each block of code. The entire pipeline can be rerun to reproduce the results over the same dataset. The full notebook is published in GitHub for reusability.

Limitations

- The dataset chosen is a single country, single cohort dataset where it includes information of Engineering students evaluated by the Colombian ICFES system. The importance of socioeconomic features and their relationship identified will not be applicable to other educational system with different circumstances.
- The models cannot track the progression of each student over time and fails to distinguish the continuously strong and weak performers.
- The models produced moderate results with R squared value ranging from 0.59 to 0.61 with almost 40 percent of variation in the Global Score which remains by the existing features. This gap in the model can be due to missing of elements like mental well-being, attendance, study habits, etc.
- The models can be built using k-fold cross validation rather than splitting a single 80: 20 train-test set; the. The hyperparameter tuning was done manually instead of techniques like Bayesian optimization. This questions the certainty of the current evaluation of the models.
- The dashboard with three pages were built for educators but its usability is not tested with any user.
- The results of the feature importance and SHAP provides the relationship of the factors that the model depends to predict the Global score but that does not explain the causation if the factors were changed.

Future work

- The unexplained variance in the results can be explained if the data includes behavioral information.
- The results of the models can be enhanced with k fold cross validation technique and systematic hyperparameter tuning.
- The explainability of the models can be made broader using techniques like LIME which would give more useful guidance for the educational institutions.
- The extension of dashboard with student specific data can be integrated into the intuitional portal.

Chapter 5 – Conclusion

This study with two main research questions to fill the problem identified in the existing studies, examined the student academic data using regression models, exploratory analysis, SHAP explainability and an interactive dashboard to create machine learning framework for predicting the academic success. The objective of developing visual analytical dashboard, identifying important elements, and building predictive models were all effectively accomplished.

The study overcome the problem of target data leakage by building leakage aware framework of four models with and evaluated R2 of 0.946 approximately. The experimental results demonstrated that Linear regression had the best performance with 0.61 R-squared metric while SHAP analysis proved that previous academic performance was the most important factor influencing the Global Score. The variation of 40% in student performance result was influenced by the social and economic factors.

The interactive dashboard improved the framework for prediction and interpretability. It made the complicated analytical results to be easily understood by educators. In order to further enhance prediction accuracy, future research will examine the framework using bigger and more varied datasets with more characteristics and validate rigorously along with systematic hyperparameter tuning.

Overall, the study proved that integrating explainable AI, machine learning models and visual analytics can offer a straightforward and efficient way for analysing and comprehending student academic performance.

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